Paylens

Affordability Copilot: Calculation Manual (v2)

Verified-payslip affordability engine · formulas, thresholds, worked examples and verified fixtures

> **Tax rules:** slabs, ₹75,000 standard deduction and ₹12,00,000 Section 87A limit as in FY 2025-26. Budget 2026 left them unchanged for FY 2026-27 (checked against public sources, Oct 2026). **All amounts in ₹.** Every number in this manual was produced by the reference engine (`engine_ref.py`) at build time, not typed by hand. Examples are for learning, not tax, legal or investment advice.

What changed from v1

v1 described seven calculators for a first-paycheck app. v2 describes one question, **"Can I afford this, and how?"**, answered from a verified payslip. The tax and IRR maths carry over; the verdict logic, the savings and essentials definitions and the data inputs are new.

v1 module	Fate in v2	What changed
1. Income Tax Engine	Kept	Adds a regime-aware March TDS shock and declared 80C/80D inputs. Unchanged slabs.
2. Safe EMI & borrowing	Kept, reframed	35% rule becomes the FOIR line and *safe_emi_ceiling*. Adds the reverse EMI formula.
3. No-Cost EMI / IRR	Kept, extended	Adds down payment, explicit cash baseline, 18% GST on the fee, EAR and nominal rate.
4. Affordability traffic light	Replaced	Worst tier across FOIR, free cash and runway. APR is a separate alert.
5. Offer decoder (7 traps)	Dropped	Only the CTC reconciliation survives, as the variable-pay input.
6. Payslip auditor	Reused	EPF check and net-pay reconciliation become validation gates on upload.
7. Life plan (50-30-20)	Reused selectively	Cushion and savings share feed the Wait & Save alternative. Essentials redefined.

1. The architectural law and request flow

The LLM never does math. Gemini reads (payslips, page text) and explains (finished numbers). Every rupee figure, rate and verdict comes from deterministic Python in `app/core`. The same inputs always give the same answer, and each answer can be re-worked by hand from this manual.

Step	Where	What happens
1	Options page	Payslip uploaded. Gemini extracts fields; Python validates them (Section 4) and builds the Verified Profile (Section 2). Profile is stored in the extension, not on a server.
2	Shop page	Content script reads price and payment options from the page. Gemini only parses the page when the DOM adapter fails.
3	POST /api/afford	Extension sends (profile, listing, selected option). Pure-Python engine returns the Verdict: true cost, impact, tier, stress tests, alternatives.
4	Card	Verdict card is rendered on the page. Optional Hindi/Kannada bullets are written by Gemini from the finished numbers only.

2. The Verified Financial Profile

Everything the engine knows about the user. **[P]** = read from the payslip, **[U]** = entered by the user, **[D]** = derived by Python.

Field	Src	Definition
gross_monthly, basic, hra, pf_employee, professional_tax, tds	P	Read from the payslip.
in_hand_verified	P/D	Payslip net pay if it reconciles (Section 4, gate G2), otherwise gross − PF − PT − TDS − other deductions. **Excludes annual variable pay.**
annual_variable_bonus	U	Non-guaranteed pay from the offer letter. Used only for the CTC check below; never counted as income.
tax_regime, declared_80c, declared_80d	U	Needed for the March shock (Section 7).
rent_monthly, is_metro	U	Rent is a fixed outflow. is_metro sets the living-cost and saving rates.
existing_emis [{amount, months_left}]	U	Active EMIs. They roll off after months_left in projections.
current_savings	U	Liquid savings.
essentials	D	rent + living_rate × in_hand. living_rate = 30% metro, 25% tier-2. **Existing EMIs are not included**, so they are never counted twice.
free_cash	D	in_hand − existing EMIs − essentials.
runway (months)	D	current_savings ÷ essentials.
safe_emi_ceiling	D	35% × in_hand − existing EMIs.
cushion_target	D	3 × essentials (EMERGENCY_CUSHION_MONTHS).
saving_pool	D	saving_rate × in_hand. saving_rate = 25% metro, 30% tier-2 (the v1 Life Plan savings share).

Worked profiles (both metro, new regime):

Quantity	Rahul	Amit
In-hand / rent / savings	₹75,000 / ₹18,000 / ₹1,50,000	₹46,800 / ₹14,000 / ₹25,000
Existing EMI	₹3,500 (3 months left)	₹8,000 (8 months left)
Essentials	18,000 + 30% × 75,000 = ₹40,500	14,000 + 30% × 46,800 = ₹28,040
Free cash	₹75,000 − 3,500 − ₹40,500 = ₹31,000	46,800 − 8,000 − 28,040 = ₹10,760
Runway today	3.70 months	0.89 months (already below 2.0: RISKY before any purchase)
Safe EMI ceiling	₹22,750	₹8,380
Cushion target (3 × essentials)	₹1,21,500	₹84,120
Saving pool (25%)	₹18,750	₹11,700

CTC reconciliation (the only v1 offer-decoder logic kept)

Real Cash Ratio = 12 × guaranteed monthly in-hand ÷ annual CTC. For Rahul: fixed pay 9,60,000 + variable 1,50,000 + employer PF 57,600 (12% × 40,000 × 12) + gratuity 23,088 (4.81% × 4,80,000) = ₹11,90,688, about 11.9 LPA. Real Cash Ratio = 9,00,000 ÷ ₹11,90,688 = 75.6%. (An exact 12 LPA would need variable pay of ₹1,59,312.)

3. Income tax engine (kept from v1)

Annual gross = monthly gross × 12. Tax is charged slab by slab; each rate applies only to the income inside its slab. Cess is 4% of tax after rebate and relief.

New regime (default)

Taxable income	Rate	Max tax from slab	Cumulative at top
Up to ₹4,00,000	0%	₹0	₹0
₹4,00,001 – ₹8,00,000	5%	₹20,000	₹20,000
₹8,00,001 – ₹12,00,000	10%	₹40,000	₹60,000
₹12,00,001 – ₹16,00,000	15%	₹60,000	₹1,20,000
₹16,00,001 – ₹20,00,000	20%	₹80,000	₹2,00,000
₹20,00,001 – ₹24,00,000	25%	₹1,00,000	₹3,00,000
Above ₹24,00,000	30%	No limit	₹3,00,000 + 30% of the excess

Taxable income = annual gross − ₹75,000. **Section 87A:** if taxable income ≤ ₹12,00,000 the rebate wipes out the tax (gross up to ₹12,75,000 is tax-free). **Marginal relief:** above ₹12,00,000 the tax cannot exceed the income above ₹12,00,000.

Annual gross	Taxable	Slab tax	After rebate / relief	+ 4% cess = total
₹9,60,000 (Rahul)	₹8,85,000	₹28,500	₹0 (rebate)	₹0.00
₹12,85,000	₹12,10,000	₹61,500	₹10,000 (relief cap)	₹10,400.00
₹15,00,000	₹14,25,000	₹93,750	₹93,750	₹97,500.00

Old regime (needed for the comparison and the March shock)

Taxable = gross − ₹50,000 − exempt HRA − professional tax − 80C (cap ₹1,50,000) − 80D (cap ₹25,000). Slabs: 0% to ₹2,50,000; 5% to ₹5,00,000; 20% to ₹10,00,000; 30% above. Section 87A applies only if taxable ≤ ₹5,00,000. Cess 4%. **Exempt HRA** = the smallest of (actual HRA, rent paid − 10% of basic, 50% of basic in metros or 40% elsewhere), all annual. Example: basic 4,80,000, HRA 2,40,000, rent 2,16,000 in a metro → min(2,40,000; 1,68,000; 2,40,000) = ₹1,68,000 exempt.

₹9,60,000 gross, no HRA claimed	New regime	Old regime
Taxable income	₹8,85,000	₹7,40,000 (after 50,000 + 2,400 + 1,50,000 + 17,600)
Slab tax	₹28,500	₹60,500
87A rebate	−₹28,500	None (taxable above ₹5,00,000)
Total tax incl. cess	₹0	₹62,920
Monthly in-hand (80,000 − 4,800 EPF − 200 PT − tax ÷ 12)	₹75,000	₹69,757

Currency check. Rates and thresholds above are unchanged for FY 2026-27. The Income-tax Act 2025 replaces the 1961 Act from FY 2026-27, so section numbers quoted here (115BAC, 87A, 10(13A), 80C, 80D) may carry new labels in official forms; the rules used are the same. **Not verified:** the list of metro cities for the 50% HRA rule (v1 lists Mumbai, Delhi, Kolkata, Chennai). Confirm before release.

4. Payslip validation gates (reused from the v1 auditor)

Run on every extracted payslip before a profile is saved. A failed gate does not block the user; it flags the field for manual confirmation.

Gate	Rule	On failure
G1 EPF base	pf_employee equals 12% × basic, or the ₹1,800 cap (12% × ₹15,000 wage ceiling).	Critical: confirm PF with the user.
G2 Net-pay reconciliation	|gross − PF − PT − TDS − other deductions − net_pay| ≤ ₹1.	Critical: use the computed value, not net_pay, until confirmed.
G3 Claimed vs computed TDS	|payslip monthly TDS − engine annual tax ÷ 12| ≤ ₹500.	Warning: show the difference; possible payroll error.
Health score (optional)	100 − 20 × critical − 10 × warnings − 5 × inconsistencies.	Display only.

Rahul passes all three: PF 4,800 = 12% × 40,000; 80,000 − 4,800 − 200 − 0 = 75,000; payslip TDS 0 = engine tax 0.

5. True cost of a payment option

Every EMI option is compared with one baseline: the **effective cash price**, what you would pay today with the instant discount. Choosing EMI forfeits that discount, which is where “No-Cost” hides its cost.

```
cash_price = list_price − cash_discount
fee_gst = processing_fee × (1 + 18%)
net_financed = cash_price − down_payment − fee_gst
net_financed = EMI/(1+r) + EMI/(1+r)^2 + … + EMI/(1+r)^n (solve for monthly r)
EAR = (1 + r)^12 − 1
nominal annual rate = 12 × r
total_outflow = down_payment + EMI × n + fee_gst
hidden_markup = total_outflow − cash_price
```

There is no closed form for r, so the engine bisects: guess a rate, compare the present value of the EMIs with net_financed, halve the range, repeat (100 iterations, far beyond the precision needed). If total_outflow ≤ cash_price the option is flagged truly zero-cost and r = 0.

Bisection trace, iPhone 16 Pro (first 6 of the iterations, cash price ₹1,14,900, 6 × ₹19,983.33, fee ₹199 + GST)

Guess for monthly r	PV of 6 EMIs	vs target ₹1,14,665.18	Next step
25.0000%	₹58,979.28	Too low	Rate must be lower
12.5000%	₹81,009.19	Too low	Rate must be lower
6.2500%	₹97,497.38	Too low	Rate must be lower
3.1250%	₹1,07,806.46	Too low	Rate must be lower
1.5625%	₹1,13,606.85	Too low	Rate must be lower
0.7812%	₹1,16,688.59	Too high	Rate must be higher

It converges to r = 1.2906% per month: EAR **16.63%**, nominal 15.49% (the figure a bank would quote). Total outflow ₹1,20,134.80, hidden markup ₹5,234.80.

Worked example from v1 (₹60,000 phone, ₹3,000 discount lost, 6 × ₹10,000, ₹999 fee)

Treatment of the fee	Net financed	Monthly r	EAR	Hidden markup
v1: no GST on fee	₹56,001.00	2.007%	26.93%	₹3,999.00
v2: 18% GST on fee (current)	₹55,821.18	2.102%	28.36%	₹4,178.82

Reverse formula (needed for alternatives): EMI = P × r ÷ (1 − (1 + r)$^{-n}$). **Not modelled:** GST that card issuers charge on the interest component of a bank EMI. It would raise every EAR here slightly, so the engine understates the cost; it is left out so the numbers stay reproducible from the listing alone.

6. Impact metrics and the verdict

```
FOIR_before = existing EMIs ÷ in_hand
FOIR_after = (existing EMIs + new EMI) ÷ in_hand
free_cash_after = in_hand − existing EMIs − new EMI − essentials
free_cash_ratio = free_cash_after ÷ in_hand
runway_after = (savings − down_payment − fee_gst) ÷ essentials
```

Runway only moves by the upfront outlay. A pure EMI option leaves it almost unchanged; cash purchases and down-payment paths consume savings. Existing EMIs are counted as active today; roll-off is applied only in projections (Section 8).

Metric	COMFORTABLE	STRETCH	RISKY
FOIR	≤ 35%	35% – 50%	> 50%
Free-cash ratio	≥ 20%	5% – 20%	< 5% (or a deficit)
Runway after	≥ 3.0 months	2.0 – 3.0 months	< 2.0 months

Verdict = the worst tier of the three metrics. This closes the v1 gap where a combination of inputs matched no colour. **APR does not set the verdict.** A ₹99 fee on a ₹3,000 item is a huge APR but a trivial cost. Instead a separate **High-Cost Financing alert** fires when EAR > 18% and hidden_markup ≥ min(₹1,500, 5% of cash price). The card shows both: a budget verdict and a cost alert.

Config constant	Default	Used for
SAFE_FOIR / STRETCH_FOIR	35% / 50%	FOIR tiers (and the stress pass line)
FC_COMFORT / FC_STRETCH	20% / 5%	Free-cash tiers
CUSHION_MONTHS / RUNWAY_STRETCH	3.0 / 2.0	Runway tiers, cushion target
LIVING_RATE (metro / tier-2)	30% / 25%	Essentials
SAVING_RATE (metro / tier-2)	25% / 30%	Saving pool for Wait & Save
GST	18%	Processing fee
APR_ALERT, ALERT_MIN_ABS, ALERT_MIN_PCT	18%, ₹1,500, 5%	High-Cost Financing alert
STRESS_INCOME_CUT	20%	Stress test A

7. Stress tests

A. Income −20%. in_hand′ = 0.8 × in_hand, essentials held fixed in rupees. Pass if FOIR′ ≤ 50% and free cash′ ≥ 0. The 20% severity is an arbitrary, tunable assumption, not a legal or statistical figure.

B. March TDS shock (regime-aware). Employers spread tax using declared deductions, then recover the extra in the last month if proofs are missing. New regime: no proofs are needed (standard deduction is automatic), so shock = ₹0 and the row is hidden. Old regime: shock = tax(no 80C/80D) − tax(declared 80C/80D), charged conservatively in one month. March free cash = free_cash_after − shock; pass if ≥ 0.

Old-regime example (Priya): gross ₹12,00,000, declared 80C ₹1,50,000 and 80D ₹25,000	Annual tax
With declarations proven (taxable 9,72,600)	₹1,11,300.80
Proofs rejected (taxable 11,47,600)	₹1,63,051.20
March shock = difference	₹51,750.40

Rahul (new regime, taxable ₹8,85,000, inside the 87A limit) has shock ₹0, so his card never shows a March row. Amit has no regime-driven shock either; his failures come from his baseline, and the card says so.

8. Alternatives

Alternatives are not advice text. Each is a concrete state (new EMI, upfront outlay, savings, existing EMIs) passed through the **same** evaluate() that scored the main option, so the card shows the real verdict of each path. Infeasible paths are hidden, never scored.

Alternative	Rule	Feasible when
Buy in cash now	upfront = cash_price; savings fall by that amount; rescored.	cash_price ≤ savings
Wait & Save	Smallest whole month m with savings(m) − cash_price ≥ cushion. savings(k) = savings(k−1) + min(saving_pool, free_cash(k)); free_cash(k) uses only EMIs still active in month k (roll-off). Rescored with the EMIs active in month m+1. Saved money = the option's hidden markup (assumes the price does not change).	Within 60 months
Down-payment boost *(hypothetical)*	d = (savings − cushion) − fee_gst; EMI′ = (cash_price − d) × r ÷ (1 − (1+r)$^{-n}$) at the option's implied r. Needs a bank or store that allows part-payment, so it is labelled as such.	d > 0
Longer tenure	EMI′ from the reverse formula at the stated rate. Only for standard EMIs with a stated rate, not No-Cost EMIs.	Rate known

Worked: Rahul, iPhone 16 Pro

Path	State after	Verdict
Buy in cash now	Savings left ₹35,100; runway 0.87 months	RISKY
Wait & Save	Saves ₹18,750/month (the pool is the binding limit, so the ₹3,500 EMI ending after month 3 does not change the answer). After 5 months savings are ₹2,43,750; after buying, runway 3.18, free cash ₹34,500. Saves the ₹5,234.80 markup.	COMFORTABLE
Down-payment boost (hypothetical)	Down ₹28,265.18, EMI′ ₹15,098.33; FOIR 24.8%, runway 3.00	COMFORTABLE

Worked: Amit, gaming laptop

Buy in cash is **not possible** (₹1,40,000 needed, ₹25,000 saved), so it is not shown. A down payment is not possible either: savings are below the cushion, so surplus is ₹0. Wait & Save becomes **“build the cushion first, then buy”**: he must reach cushion ₹84,120 + price ₹1,40,000, i.e. savings of ₹2,24,120. With free cash ₹10,760 while the bike EMI lasts (8 months), then the pool of ₹11,700, this takes **18 months**. Afterwards: runway 3.14, free cash ₹18,760 (40%), verdict COMFORTABLE.

9. Verified fixtures

These are the test fixtures for `test_engine.py` and the demo. Tier abbreviations: COMF = comfortable, STRE = stretch, RISK = risky.

Quantity	Sony WH-1000XM5 (Rahul)	iPhone 16 Pro (Rahul)	Gaming laptop (Amit)
List price / cash discount	₹29,990 / ₹2,000	₹1,19,900 / ₹5,000	₹1,45,000 / ₹5,000
Effective cash price	₹27,990.00	₹1,14,900.00	₹1,40,000.00
Option	6 × ₹4,998.33, fee ₹199 + GST	6 × ₹19,983.33, fee ₹199 + GST	12 × ₹13,160.00, fee ₹999 + GST
Total outflow	₹30,224.80	₹1,20,134.80	₹1,59,098.82
Hidden markup	₹2,234.80	₹5,234.80	₹19,098.82
EAR / nominal annual	30.74% / 27.10%	16.63% / 15.49%	27.44% / 24.49%
FOIR before → after	4.7% → 11.3%	4.7% → 31.3%	17.1% → 45.2%
Free cash after (ratio)	₹26,001.67 (34.7%)	₹11,016.67 (14.7%)	-₹2,400.00 (-5.1%)
Runway after	3.70 months	3.70 months	0.85 months
Tiers FOIR / cash / runway	COMF / COMF / COMF	COMF / STRE / COMF	STRE / RISK / RISK
Budget verdict	COMFORTABLE	STRETCH	RISKY
High-Cost Financing alert	YES	No	YES
Stress A (income −20%)	Pass: FOIR 14.2%, free cash ₹11,001.67	Fail: FOIR 39.1%, free cash -₹3,983.33	Fail: FOIR 56.5%, free cash -₹11,760.00
Buy in cash now	COMFORTABLE: runway 3.01	RISKY: runway 0.87	Not possible
Wait & Save	Buy now (0 months)	5 months → COMFORTABLE	18 months → COMFORTABLE

Differences from the pivot plan's numbers

1. **Wait & Save for the iPhone is 5 months, not 3.** The plan saved all free cash; v2 saves only the saving pool (25% of in-hand).
2. **Runway after** now subtracts the upfront fee, so Amit's is 0.85, not 0.89.
3. **Amit's Wait & Save is 18 months** because it includes building the cushion first.
4. **Rahul's variable pay is ₹1,50,000** (≈ 11.9 LPA), not ₹1,82,400; the earlier figure forgot gratuity.
5. The down-payment alternative is hypothetical and kept out of the must-ship set.

10. Quick reference

Topic	Formula
Annual gross	monthly gross × 12
New / old taxable	gross − 75,000 / gross − 50,000 − HRA − PT − min(80C, 1.5L) − min(80D, 25,000)
Tax	slab tax; 87A and marginal relief; × 1.04 cess
Essentials	rent + living_rate × in_hand (30% metro, 25% tier-2)
Free cash / runway	in_hand − EMIs − essentials / savings ÷ essentials
Safe EMI / cushion / saving pool	35% × in_hand − EMIs / 3 × essentials / saving_rate × in_hand
True cost	net = cash − down − fee × 1.18; solve PV(EMI, r, n) = net; EAR = $(1+r)^{12} - 1$
Reverse EMI	$P \times r \div (1 - (1+r)^{-n})$
FOIR / free-cash ratio / runway	(EMIs + new) ÷ in_hand / free cash ÷ in_hand / (savings − upfront) ÷ essentials
Verdict	worst tier of FOIR, free-cash ratio and runway
High-cost alert	EAR > 18% and markup ≥ min(₹1,500, 5% × cash price)
March shock	0 under new regime; tax(no proofs) − tax(declared) under old
Wait & Save	min m: savings(m) − price ≥ cushion; monthly add = min(pool, free cash)
Rounding	Money to 2 decimals internally, ₹ whole in the UI; months rounded up; EAR shown to 2 decimals

11. Assumptions, limitations and release checklist

#	Item	Status
1	Essentials are a heuristic (rent + 30% or 25% of in-hand). It should be shown in the UI as an assumption and editable.	Assumption
2	Saving pool rates (25% / 30%) come from the v1 50-30-20 calibration. They decide Wait & Save length. Tunable.	Assumption
3	Stress A severity (−20%) is arbitrary.	Assumption
4	GST on the interest part of bank EMIs is not modelled; EAR is therefore a slight understatement.	Known gap
5	Wait & Save assumes the price and the discount stay the same while waiting.	Known gap
6	EAR is shown with the nominal rate beside it, because banks quote nominal.	Rule
7	OCR accuracy figures from the benchmark come from synthetic payslips; say so in the pitch.	Disclosure
8	Verify the HRA metro-city list and the Income-tax Act 2025 section labels before release.	Open
9	Update pay_period fixtures (they still say March 2026) and re-run the fixtures after any config change.	Open
10	scripts/compute_fixtures.py must import from app/core, so this manual and the code cannot drift apart.	Open